K-Means

Unsupervised Machine Learning
Clustering Algorithm

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Course Topics

- ✔ What is K-Means?
- ✔ How the algorithm works
- ✔ Pros and Cons
- ✔ Applications
- ✔ Modeling kmeans

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What is K-Means ?

- *K*-means is a type of unsupervised machine learning algorithm
- Used for clustering unlabelled data
- Looks to find groups in the data by minimize the distance between the data points and their respective cluster centroid, with the number of groups represented by the variable *K*.
- The algorithm works iteratively to assign each data point to one of *K* groups based on the features that are provided.
- Data points are clustered based on feature similarity.

Supervised Learning

X_1	X_2	X_3	X_p	Y

Target

Un-Supervised Learning

X_1	X_2	X_3	X_p	Y

No Target

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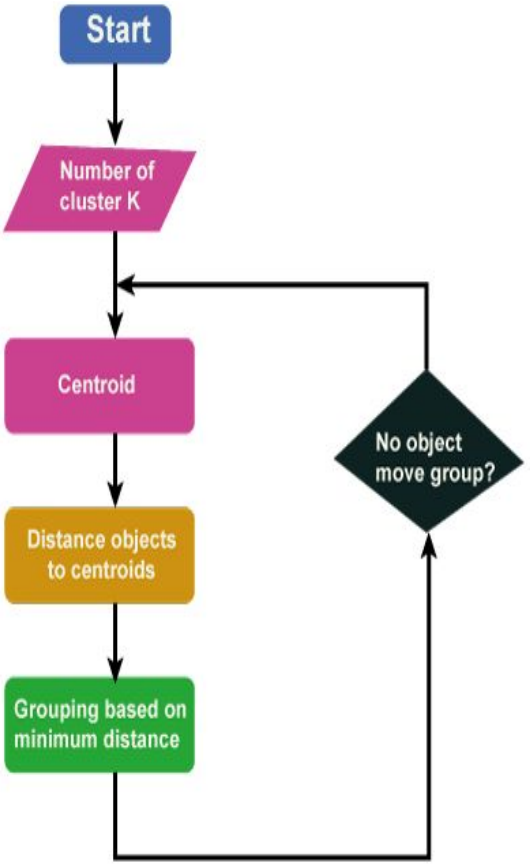
How K-Means works

1. Initialize no. of clusters (K).
2. Randomly assign k no. of centroids into a dataset.
3. Calculate the distance between each datapoint to all the centroids. Assign the datapoints to the nearest centroid so as to form k clusters.
4. Compute the new centroids of each cluster based on the data it contains, shifting from the older centroids.
5. Repeat steps 3 and 4 until the mean doesn't change.



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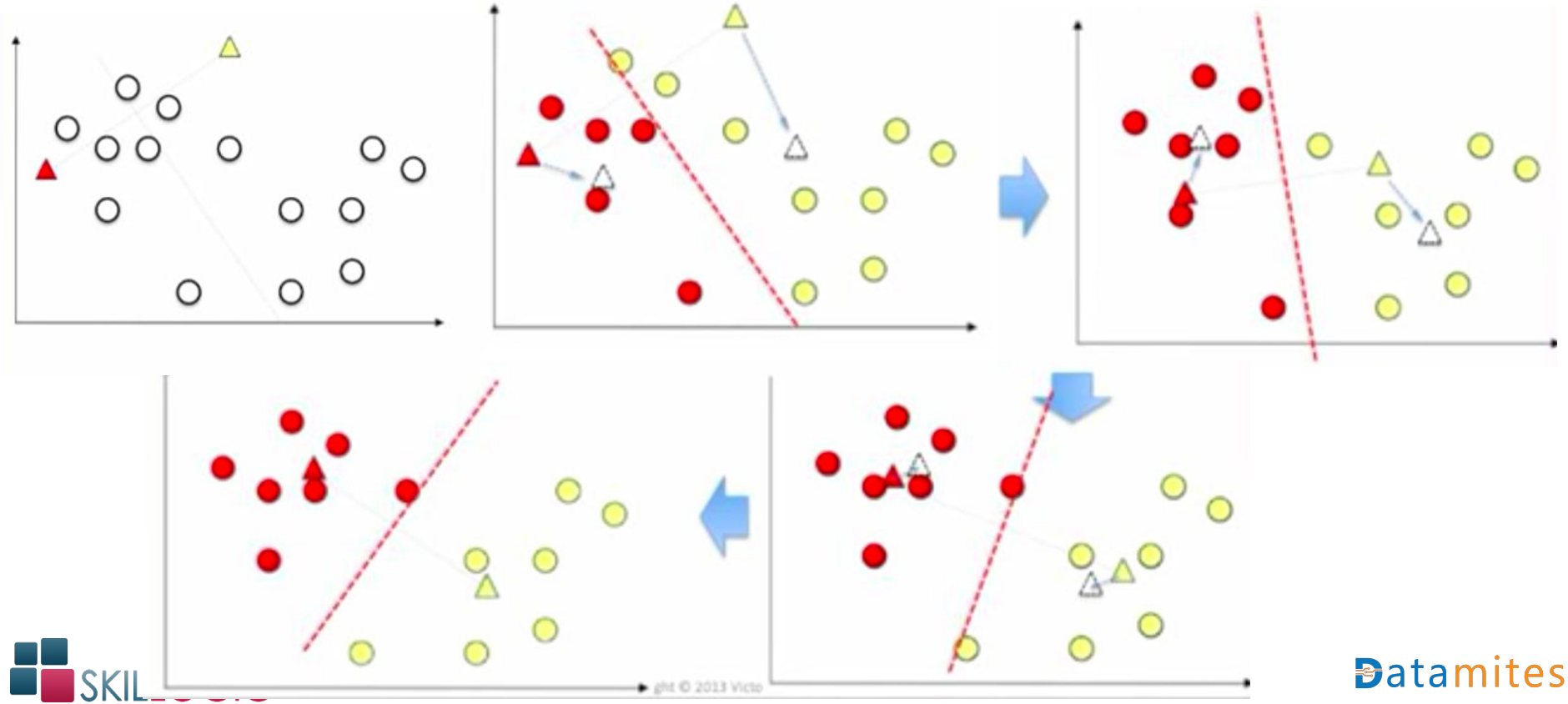
How K-Means works?



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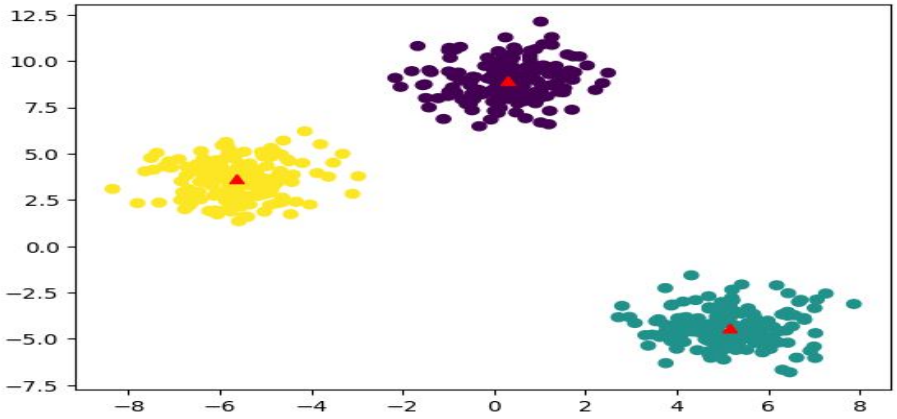
Pros of K-Means

- Practically work well even if some assumptions are broken.
- Simple, easy to implement.
- Easy to interpret the clustering results.
- Fast and efficient in terms of computational cost.



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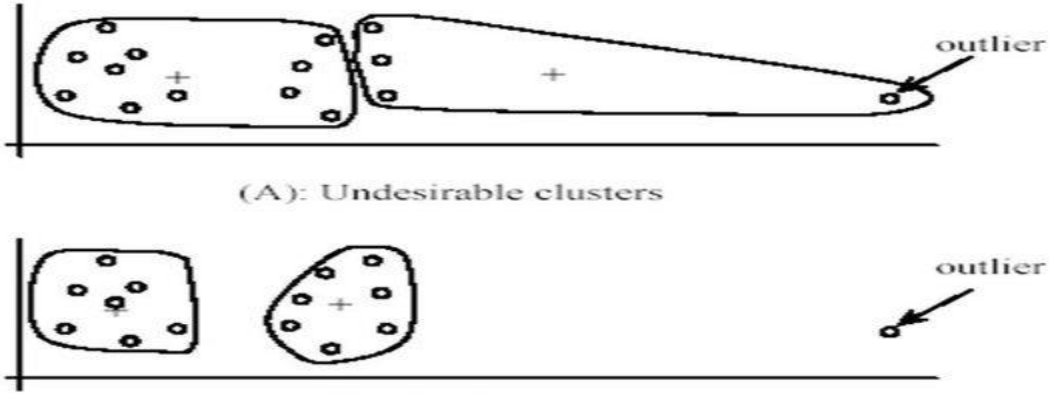
Cons of K-Means

- **Uniform effect:** often produce clusters with relatively uniform size even if the input data have different cluster size.
- **Different densities:** may work poorly with clusters with different densities but spherical shape.
- Sensitive to outliers
- K value needs to be known before K-means clustering



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Applications of K-Means

- Market segmentation
- Computer vision
- Agriculture
- Feature learning
- Recommendation Systems

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Modeling K-Means

Clustering in action

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